

YUMA, AZ, USA

WMO: 722800

Lat: 32.650N

Lon: 114.600W

Elev: 207

StdP: 14.59

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	42.3	44.8	3.9	6.8	60.8	8.3	8.5	62.5	24.7	61.0	21.0	60.0	4.2	30	0.452

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	111.0	72.3	108.9	72.2	107.6	71.6	79.3	96.0	78.1	95.1	77.2	94.7	8.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	133.1	86.7	73.3	124.8	87.5	72.1	119.8	88.1	42.9	96.0	41.7	94.2	40.7	94.1	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.5	18.8	16.8	DB	36.0	116.4	3.0	2.4	33.8	118.1	32.1	119.5	30.4	120.8	28.2	122.6
			WB	29.4	82.1	2.9	2.6	27.3	84.0	25.6	85.5	24.0	87.0	21.9	88.9

Monthly Climatic Design Conditions

		Annual														
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.3	59.4	61.9	68.1	73.1	80.5	89.5	95.0	95.0	89.5	77.9	66.6	57.9	(6)
(7)		DBStd	14.27	5.01	5.77	6.42	6.67	6.53	5.45	3.49	3.47	5.14	6.45	6.08	4.97	(7)
(8)		HDD50	7	3	1	0	0	0	0	0	0	0	0	0	3	(8)
(9)		HDD65	637	181	118	43	14	1	0	0	0	0	3	52	225	(9)
(10)		CDD50	9604	295	335	562	694	946	1186	1395	1395	1186	864	498	248	(10)
(11)		CDD65	4759	8	32	140	258	481	736	930	930	736	402	101	5	(11)
(12)		CDH74	70129	98	362	1610	3035	6370	11411	15363	15252	11075	4580	915	58	(12)
(13)		CDH80	45104	6	75	637	1469	3719	7633	10934	10818	7164	2366	280	3	(13)
(14)	Wind	WSAvg	7.2	6.8	6.5	6.8	7.9	7.8	7.8	8.7	8.2	6.8	6.2	6.2	6.7	(14)
(15)	Precipitation	PrecAvg	2.9	0.4	0.2	0.2	0.2	0.0	0.0	0.2	0.6	0.3	0.2	0.2	0.4	(15)
(16)		PrecMax	6.2	2.1	1.2	1.6	1.2	0.4	0.3	2.1	3.4	2.5	2.5	1.7	2.1	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	1.5	0.4	0.3	0.4	0.3	0.1	0.0	0.4	0.9	0.6	0.5	0.4	0.6	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.4	86.3	94.6	100.1	107.9	113.1	114.8	113.3	109.3	102.2	90.5	79.2	(19)
(20)			MCWB	55.5	57.4	62.1	63.5	69.3	68.9	73.6	72.9	71.2	67.9	61.0	57.1	(20)
(21)		2%	DB	76.7	82.2	90.4	96.3	102.4	109.3	111.0	110.7	107.2	98.7	87.0	75.1	(21)
(22)			MCWB	53.6	56.8	60.5	62.3	65.6	69.9	72.9	73.3	70.8	66.8	59.9	54.6	(22)
(23)		5%	DB	73.4	78.9	87.0	91.7	99.6	107.4	108.8	108.6	104.7	94.9	83.0	72.1	(23)
(24)			MCWB	52.9	55.7	58.7	60.6	65.2	68.9	72.6	73.4	70.7	64.4	59.2	54.0	(24)
(25)		10%	DB	71.3	74.8	83.2	88.5	96.8	104.5	106.8	106.3	101.9	91.5	79.5	68.7	(25)
(26)			MCWB	52.5	54.1	57.6	59.3	63.8	67.9	72.3	73.3	71.1	63.7	57.6	52.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.4	62.3	66.4	67.8	74.3	76.8	81.4	81.1	80.3	74.9	67.3	61.8	(27)
(28)			MCDWB	71.4	77.7	89.6	94.3	100.9	103.1	102.9	97.2	92.9	88.7	82.0	72.5	(28)
(29)		2%	WB	58.8	59.9	62.2	64.3	69.3	74.7	78.4	79.6	78.4	72.4	63.7	58.9	(29)
(30)			MCDWB	68.4	74.5	84.4	90.7	97.2	99.9	98.4	95.6	92.1	88.3	77.9	69.8	(30)
(31)		5%	WB	56.7	58.1	60.3	62.2	66.7	72.7	77.3	78.4	77.3	70.0	61.4	56.9	(31)
(32)			MCDWB	67.0	71.7	82.3	88.0	94.8	98.2	96.3	95.2	91.3	86.1	76.7	67.4	(32)
(33)		10%	WB	54.5	56.2	58.7	60.3	64.7	70.7	76.3	77.5	76.0	67.8	59.4	54.4	(33)
(34)			MCDWB	66.2	70.7	80.1	84.9	92.0	97.3	96.0	94.9	91.9	84.9	76.6	65.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	21.3	22.3	24.5	25.3	26.4	27.2	23.1	22.0	22.6	23.5	22.4	20.4	(35)
(36)			MCDBR	24.7	27.2	29.6	29.7	29.7	29.8	25.8	24.6	25.5	26.8	26.8	24.3	(36)
(37)			MCWBR	11.8	11.8	11.7	11.9	11.3	11.0	8.8	7.9	8.5	10.7	11.7	12.5	(37)
(38)		5% WB	MCDWB	19.7	22.3	26.7	27.6	27.7	26.6	21.5	20.5	20.2	21.6	22.8	20.7	(38)
(39)			MCWBR	11.9	10.7	11.7	12.1	12.9	11.1	8.5	8.0	9.5	10.5	12.5	13.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.290	0.294	0.315	0.324	0.337	0.342	0.386	0.383	0.347	0.322	0.297	0.290	(40)	
(41)		taud	2.495	2.502	2.419	2.403	2.383	2.379	2.370	2.368	2.431	2.454	2.498	2.512	(41)	
(42)		Ebn at Noon	297	307	306	306	300	297	284	284	292	292	292	291	(42)	
(43)		Edh at Noon	27	30	35	37	39	39	39	38	34	31	27	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1133	1446	1904	2327	2584	2670	2415	2252	1988	1613	1229	1023	(44)	
(45)		RadStd	52	90	60	52	59	91	83	85	65	57	51	52	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	-3.91	N/A	-0.79	N/A	N/A	N/A	N/A
(47)	Regional (8 neighbors)	+0.59	N/A	-2.90	N/A	-0.65	N/A	N/A	+232	+166

Nomenclature: See separate page